

Sensitivity study: $B^0 \rightarrow K^{*0}(K^+\pi^-)\ell^+\ell^-$ with a $J/\psi K^{*0}$ peaking background

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This study uses only publicly available tools and publications. No Belle II internal materials — data, MC, internal notes, or the software framework — are used. Everything is open-source and runs on a laptop.

Abstract

Using existing fast-simulation ntuples (10000 generated $B^0 \rightarrow K^{*0}\ell^+\ell^-$ signal events and 5000 generated $B^0 \rightarrow J/\psi(\rightarrow \ell^+\ell^-)K^{*0}$ background events, split evenly between e^+e^- and $\mu^+\mu^-$), we apply the Belle/Belle II $B \rightarrow K^*(892)\ell^+\ell^-$ selection: a $K\pi$ mass window (0.896 ± 0.075 GeV), $M_{bc} > 5.27$ GeV, an asymmetric per-flavor ΔE window, and an asymmetric charmonium veto in $m_{\ell\ell}$. Before the veto the $J/\psi K^{*0}$ control region is essentially pure background (20645 e^+e^- and 19132 $\mu^+\mu^-$ events at 1 ab^{-1} , purity 99.89%); after the veto the leakage drops to 43.2 events (e^+e^-) and consistent with zero, 68%-CL upper limit ~ 25 events ($\mu^+\mu^-$), a signal-efficiency step of 92.5% versus a background rejection of 99.8–100%. At 1 ab^{-1} the projected yields are 285.7 e^+e^- and 251.8 $\mu^+\mu^-$ signal candidates, giving $\sim 6.3\%$ relative statistical precision per flavor and 4.5% combined. Continuum and generic- $B\bar{B}$ combinatorial background are not modeled in this study.

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1 Introduction

$B^0 \rightarrow K^{*0} \ell^+ \ell^-$ ($\ell = e, \mu$) is a flavor-changing-neutral-current $b \rightarrow s \ell^+ \ell^-$ transition, forbidden at tree level in the Standard Model and sensitive to new heavy particles in the loop. Belle first measured its branching fraction and forward-backward asymmetry using the beam-energy-constrained mass M_{bc} , the energy difference ΔE , and a $K\pi$ invariant-mass window around the $K^*(892)$, with asymmetric charmonium (J/ψ , $\psi(2S)$) vetoes in the dilepton mass that are wider on the low side for electrons because of bremsstrahlung (Wei et al. [Belle], PRL 103, 171801 (2009) [1]). The vetoed $J/\psi K^*$ sample itself served as the control channel for efficiencies and PID. The most recent Belle II measurement of this same final state (arXiv:2206.05946 [2]) uses the identical strategy and quotes explicit numerical veto windows, J/ψ excluded at $M(\mu^+ \mu^-) \notin [2.946, 3.176]$ GeV and $M(e^+ e^-) \notin [2.846, 3.176]$ GeV, which this note adopts as the starting point and validates on its own simulation.

This note reproduces that selection logic on a fast-simulation sample to estimate the signal yield and statistical precision achievable with 1 ab^{-1} , and to quantify the size and control-region purity of the residual $J/\psi K^{*0}$ peaking background after the veto. Continuum and generic- $B\bar{B}$ combinatorial background are explicitly **not** modeled in this study (no such samples were generated); this is stated wherever it matters and discussed in Section 7.

2 Data samples and analysis tools

Samples were **not regenerated** for this study; two existing ntuples (produced by the script `make_ntuple_exclusive.py`, one row per reconstructed B candidate) were used directly.

Sample	Process	Generator model	Events	Flavor split	Seeds
sig_kstill.root	$B^0 \rightarrow K^{*0}(K^+ \pi^-) \ell^+ \ell^-$	EvtGen BTOSLLBALL	10000	50% ee , 50% $\mu\mu$	32 / 33
bkg_jpsikst.root	$B^0 \rightarrow J/\psi(\rightarrow \ell\ell) K^{*0}(K^+ \pi^-)$	EvtGen ($J/\psi K^* + J/\psi \rightarrow \ell\ell$)	5000	50% ee , 50% $\mu\mu$	34 / 35

Table 1: MC samples used in this study.

Relevant branches: `m_visible` (reconstructed $m(K\pi)$, i.e. the K^* candidate mass), `mbc`, `delta_e`, `m_ll` (reconstructed dilepton mass), `lep_flavor` ($11 = e$, $13 = \mu$). These ntuples carry no `true_mode` branch (unlike the `make_ntuple` output used in other notes of this framework): truth information here is simply which file a candidate came from.

The events were passed through the framework’s fast detector simulation (track/cluster smearing, lepton and hadron PID) at ntuple-production time; only ~ 44 – 54% of generated events per flavor yield a reconstructed candidate row in the ntuple (Section 4, “reconstructed” cutflow stage) — this already includes basic reconstruction/acceptance losses.

Normalization to 1 ab^{-1} . The number of available $B^0/\bar{B}^0$ mesons is

$$N(B^0 \text{ avail}) = 2 \times f_{00} \times \sigma(B\bar{B}) \times L = 2 \times 0.486 \times 1.1 \text{ nb} \times 10^9 \text{ nb}^{-1} = 1.0692 \times 10^9$$

using $f_{00} = 0.486$, $\sigma(B\bar{B}) = 1.1 \text{ nb}$, and $L = 1 \text{ ab}^{-1} = 10^9 \text{ nb}^{-1}$. Per-event luminosity weights (expected events at 1 ab^{-1} per generated MC event, by flavor) are

$$w = \frac{N(B^0 \text{ avail}) \times \mathcal{B}(\text{mode}) \times \mathcal{B}(K^{*0} \rightarrow K^+ \pi^-)}{N_{\text{generated}}(\text{flavor})}$$

with $\mathcal{B}(K^{*0} ee) = 1.03 \times 10^{-6}$, $\mathcal{B}(K^{*0} \mu\mu) = 1.05 \times 10^{-6}$, $\mathcal{B}(K^{*0} \rightarrow K^+ \pi^-) = 2/3$, $\mathcal{B}(J/\psi K^{*0}) = 1.27 \times 10^{-3}$, $\mathcal{B}(J/\psi \rightarrow \ell\ell) = 0.0597$ (same for e and μ), and $N_{\text{generated}} = 5000$ per signal flavor, 2500 per background flavor:

w (events / 1 ab^{-1} per generated MC event)	
signal, e^+e^-	0.1468
signal, $\mu^+\mu^-$	0.1497
$J/\psi K^{*0}$, e^+e^-	21.618
$J/\psi K^{*0}$, $\mu^+\mu^-$	21.618

Table 2: Per-event luminosity weights.

The huge weight of the $J/\psi K^{*0}$ sample reflects $\mathcal{B}(J/\psi K^{*0})$ being $\sim 1000\times$ larger than the signal BF: the charmonium veto (Section 3.4) must reject this background very efficiently.

All figures were produced with `plot_stacked` (luminosity-weighted, truth-split by sample and, where relevant, by reconstructed lepton flavor via a `lep_flavor` selection). All cut values were fixed with the `scan_cut` tool, scanning one variable at a time on top of the preceding cuts, using unweighted MC counts as the figure of merit $S/\sqrt{S+B}$ (a shape-level proxy; the weighted, physical yields are computed separately in Section 6).

3 Event selection

Four sequential cuts are applied, in the same order and with the same physics motivation as Wei et al. [Belle] [1] and the Belle II $K^*(892)\ell\ell$ measurement [2].

3.1 K^* mass window (m_{visible})

Targets: wrong-mass $K\pi$ combinations under the $K^*(892)$ lineshape (natural width 47 MeV) folded with detector resolution. In this simplified simulation both the signal and the $J/\psi K^{*0}$ background contain a genuine $K^*(892)$, so this cut is a resolution/acceptance cut rather than a signal/background discriminant — as confirmed by the scan, where the FOM falls monotonically as the window is tightened for both samples equally (`scan_cut` on `m_visible`, symmetric window):

window (GeV)	signal eff.	background eff.
± 0.05	74.8%	75.7%
± 0.075 (chosen)	83.6%	83.7%
± 0.10	87.8%	88.4%

Table 3: Scan of the symmetric K^* mass window.

Window chosen: **$0.821 < m(K\pi) < 0.971 \text{ GeV}$** ($0.896 \pm 0.075 \text{ GeV}$), retaining $\sim 84\%$ of both samples while removing the far tails where a real combinatorial background (not modeled here) would dominate. Figure 1 shows the distribution with cut lines at the window edges.

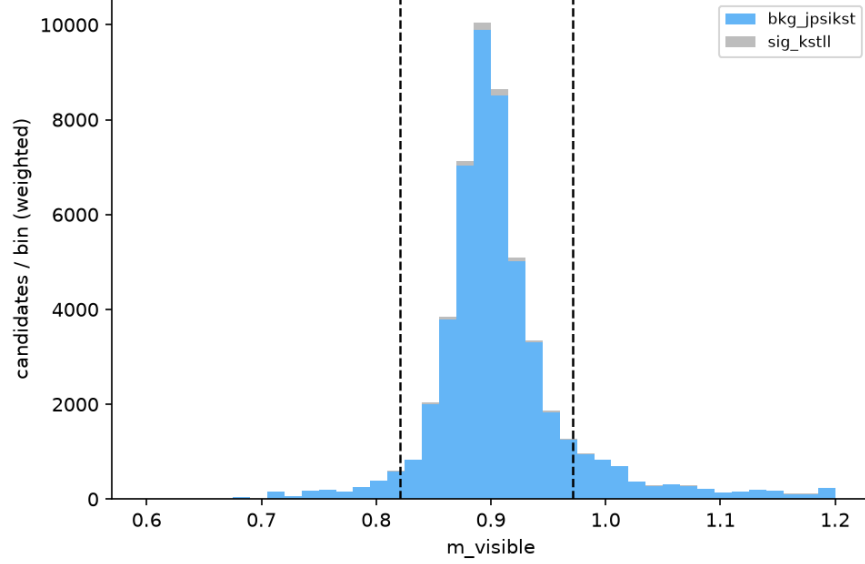


Figure 1: Reconstructed $K\pi$ invariant mass (K^* candidate mass) before any selection, stacked by sample (luminosity-weighted); the vertical lines mark the chosen $0.821 < m(K\pi) < 0.971$ GeV window.

3.2 Beam-energy-constrained mass (M_{bc})

Targets: combinatorial/continuum background populating low M_{bc} (not modeled here) and mis-reconstructed candidates; real signal peaks at the beam energy. Scan on top of the K^* window (`scan_cut`, `mbc`, direction “>”):

$M_{bc} > (\text{GeV})$	S	B	$S/\sqrt{S+B}$
5.24	4103	1960	52.7
5.26	4048	1931	52.4
5.27 (chosen)	3999	1900	52.1
5.275	3953	1875	51.8
5.28	860	425	24.0

Table 4: Scan of the M_{bc} requirement on top of the K^* window.

The sharp drop above 5.28 GeV marks the edge of the M_{bc} peak itself (beam-energy edge, ~ 5.29 GeV, smeared by detector resolution); cutting there would remove signal, not background. $M_{bc} > 5.27$ GeV is chosen, following Wei et al., retaining 96% of the K^* -window sample. Figure 2 shows the distribution with the cut line at 5.27 GeV.

3.3 Energy difference (ΔE), per lepton flavor

Targets: partially reconstructed or extra-energy background; for electrons the signal itself develops a low-side tail from final-state bremsstrahlung not fully recovered by clustering, motivating an asymmetric, wider low-side window (as in Wei et al.). Scans on top of the K^*+M_{bc} selection, separately per flavor (`scan_cut`, `delta_e`):

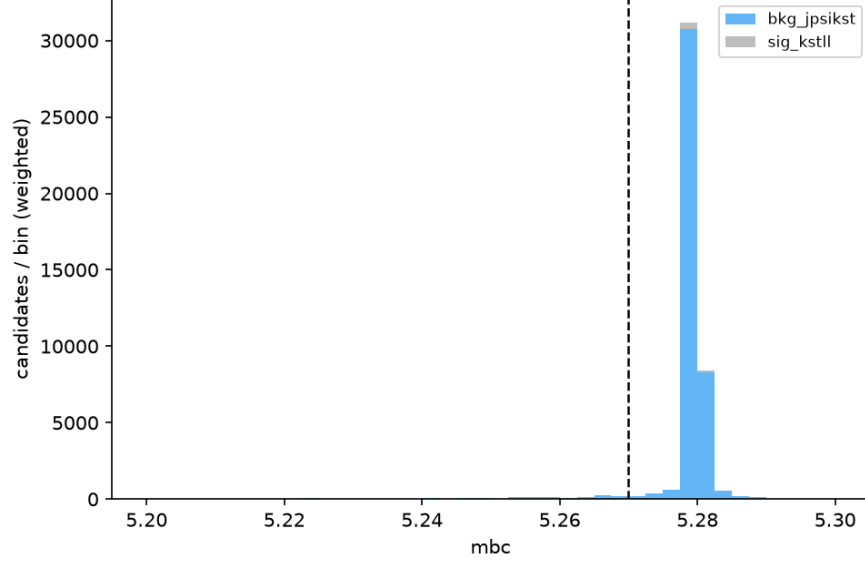


Figure 2: Beam-energy-constrained mass M_{bc} after the K^* mass window, stacked by sample (luminosity-weighted); the vertical line marks the $M_{bc} > 5.27$ GeV requirement.

Electron channel (lower edge, direction “>”): FOM rises monotonically as the window is loosened and plateaus around -0.25 to -0.30 GeV (38.0–38.2), while the upper edge saturates at $+0.05$ GeV (S stops increasing beyond that). Chosen: $-0.25 < \Delta E < 0.05$ GeV.

Muon channel: the same scan shows a much smaller low-side tail (FOM 35.2–35.3 essentially flat from -0.10 to -0.30 GeV), consistent with the absence of bremsstrahlung; the upper edge again saturates at $+0.05$ GeV. Chosen: $-0.10 < \Delta E < 0.05$ GeV, a narrower low side than for electrons, reproducing the qualitative asymmetry reported by Wei et al.

Figures 3 and 4 show the distributions with cut lines at the chosen edges.

3.4 Charmonium (J/ψ) veto in $m_{\ell\ell}$, asymmetric per flavor

Targets: $B^0 \rightarrow J/\psi(\rightarrow \ell^+\ell^-)K^{*0}$, the dominant peaking background, $\sim 1000\times$ the signal rate before any veto. The trade-off scan (`scan_cut`, `m_11`, on top of $K^* + M_{bc} + \Delta E$, per flavor) brackets the peak from below and above:

Electron channel, lower edge (keep $m_{\ell\ell} < \text{threshold}$): background stays at 0–2 events up to 2.9 GeV then rises quickly (11 at 2.95, 30 at 3.0). Upper edge (keep $m_{\ell\ell} > \text{threshold}$): background jumps from 297 (> 3.10 , inside the radiative tail region) to 4 (> 3.13) to 0 (≥ 3.15).

Muon channel, lower edge: background is 0 up to 2.98 GeV, then rises (7 at 3.02, 26 at 3.05). Upper edge: background jumps from 316 (> 3.10) to 1 (> 3.13) to 0 (≥ 3.15).

Both flavors therefore have a “safe” plateau (background ~ 0 , signal near its local maximum) that brackets the Belle II window values; these are adopted as the final veto, confirmed rather than re-derived by the scan:

Figures 5 and 6 show the dilepton-mass distributions with the veto windows.

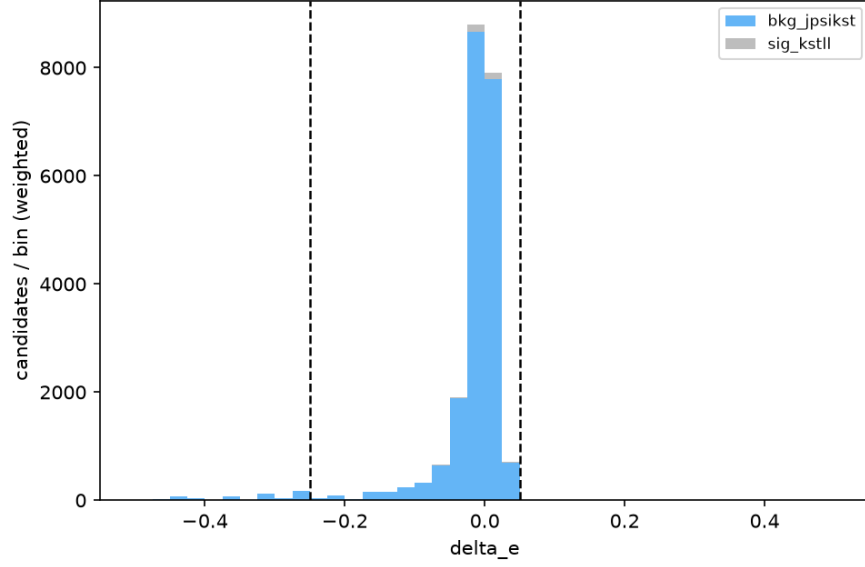


Figure 3: Energy difference ΔE in the electron channel after the K^* and M_{bc} requirements, stacked by sample (luminosity-weighted); the vertical lines mark the asymmetric $-0.25 < \Delta E < 0.05$ GeV window, wider on the low side because of bremsstrahlung.

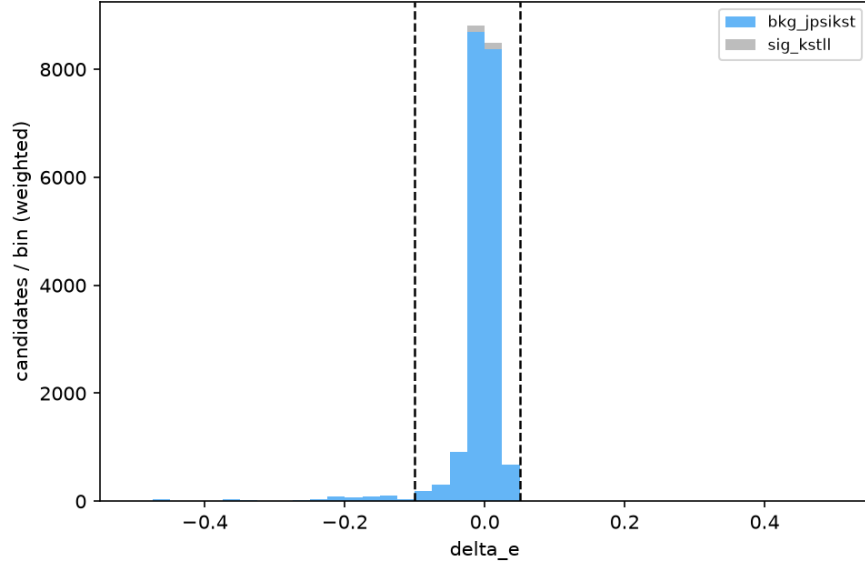


Figure 4: Energy difference ΔE in the muon channel after the K^* and M_{bc} requirements, stacked by sample (luminosity-weighted); the vertical lines mark the narrower $-0.10 < \Delta E < 0.05$ GeV window.

flavor	veto window (excluded)
e^+e^-	2.846 – 3.176 GeV
$\mu^+\mu^-$	2.946 – 3.176 GeV

Table 5: Adopted charmonium veto windows in $m_{\ell\ell}$ (Belle II, arXiv:2206.05946 [2]).

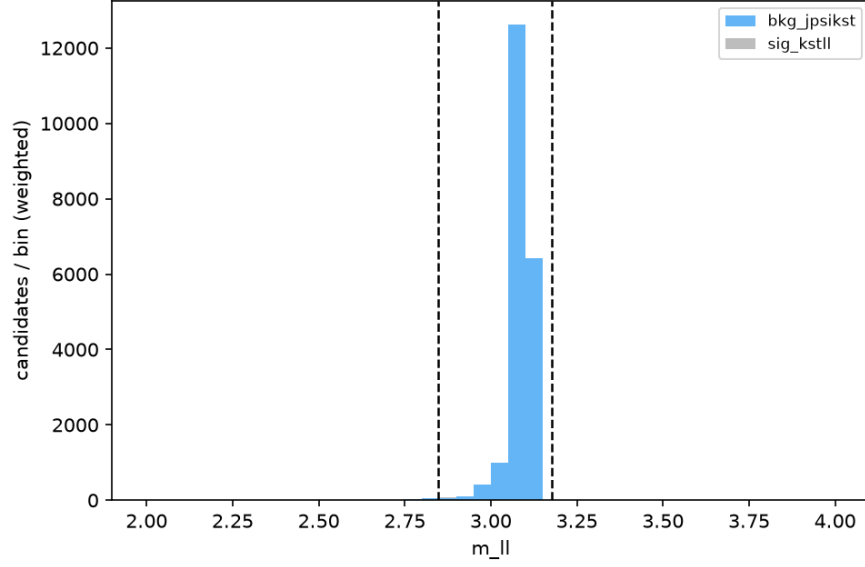


Figure 5: Dilepton mass $m_{\ell\ell}$ in the electron channel after the K^* , M_{bc} and ΔE requirements, stacked by sample (luminosity-weighted); the vertical lines mark the asymmetric J/ψ veto window $[2.846, 3.176]$ GeV, wider on the low side because of the bremsstrahlung tail.

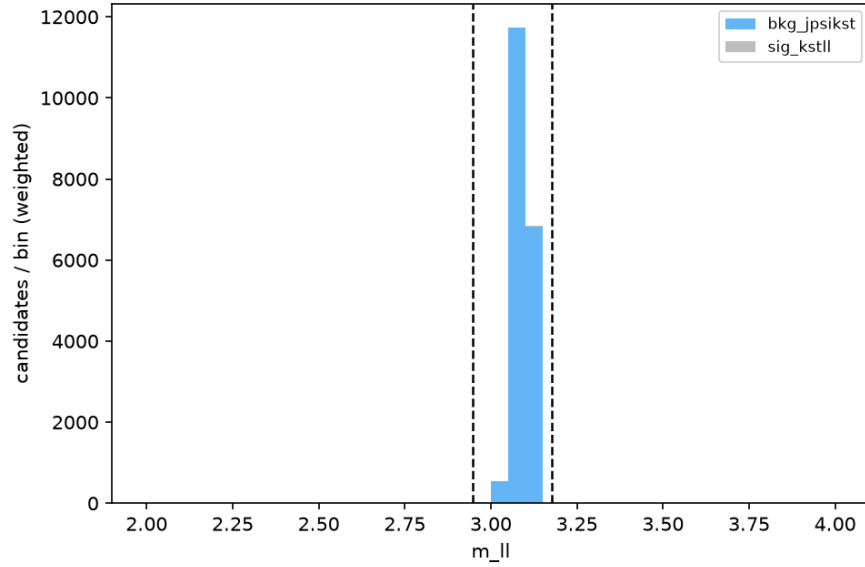


Figure 6: Dilepton mass $m_{\ell\ell}$ in the muon channel after the K^* , M_{bc} and ΔE requirements, stacked by sample (luminosity-weighted); the vertical lines mark the J/ψ veto window $[2.946, 3.176]$ GeV.

4 Full cutflow (per flavor, per sample)

Efficiencies are cumulative relative to the number of **generated** events for that flavor (5000 for signal, 2500 for $J/\psi K^{*0}$ background); “reconstructed” is the fast-sim reconstruction/acceptance stage already built into the ntuple.

stage	N	step eff.	cum. eff.
generated	5000	—	100.0%
reconstructed	2701	54.0%	54.0%
K^* window	2268	84.0%	45.4%
$M_{bc} > 5.27$	2138	94.3%	42.8%
ΔE window	2104	98.4%	42.1%
J/ψ veto	1946	92.5%	$38.9 \pm 0.7\%$

Table 6: Signal cutflow, $B^0 \rightarrow K^{*0} e^+ e^-$ (5000 generated).

stage	N	step eff.	cum. eff.
generated	5000	—	100.0%
reconstructed	2292	45.8%	45.8%
K^* window	1907	83.2%	38.1%
$M_{bc} > 5.27$	1861	97.6%	37.2%
ΔE window	1819	97.7%	36.4%
J/ψ veto	1682	92.5%	$33.6 \pm 0.7\%$

Table 7: Signal cutflow, $B^0 \rightarrow K^{*0} \mu^+ \mu^-$ (5000 generated).

stage	N	step eff.	cum. eff.
generated	2500	—	100.0%
reconstructed	1295	51.8%	51.8%
K^* window	1068	82.5%	42.7%
$M_{bc} > 5.27$	986	92.3%	39.4%
ΔE window	957	97.1%	38.3%
J/ψ veto	2	0.2%	$0.08 \pm 0.06\%$

Table 8: Background cutflow, $B^0 \rightarrow J/\psi(\rightarrow e^+ e^-) K^{*0}$ (2500 generated).

The veto is the only cut that discriminates signal from background (step efficiency 92.5% for signal vs. 0.2%/0% for the J/ψ background); all preceding cuts have essentially equal efficiency on both samples, as expected since they share the same K^* and B kinematics.

5 Background estimation

The only background modeled in this study is the residual leakage of $B^0 \rightarrow J/\psi(\rightarrow \ell^+ \ell^-) K^{*0}$ past the charmonium veto (Section 3.4); this is the dominant peaking background identified by

stage	N	step eff.	cum. eff.
generated	2500	—	100.0%
reconstructed	1101	44.0%	44.0%
K^* window	938	85.2%	37.5%
$M_{bc} > 5.27$	914	97.4%	36.6%
ΔE window	885	96.8%	35.4%
J/ψ veto	0	0.0%	0.0% (0/2500)

Table 9: Background cutflow, $B^0 \rightarrow J/\psi(\rightarrow \mu^+\mu^-)K^{*0}$ (2500 generated).

Wei et al. and Belle II for this final state. Continuum $e^+e^- \rightarrow q\bar{q}$ and generic- $B\bar{B}$ combinatorial background are **not modeled** (no such MC was generated for this study) — see Discussion.

Control region. The J/ψ veto window itself (before the veto is applied) is a natural control region for this background, exactly as used by Wei et al. and Belle II to calibrate efficiencies. Taking the candidates in that window after the $K^*+M_{bc}+\Delta E$ cuts and normalizing to 1 ab^{-1} :

flavor	$J/\psi K^{*0}$ yield in window	signal leakage in window	purity
e^+e^-	20645	23.2	99.89%
$\mu^+\mu^-$	19132	20.5	99.89%

Table 10: J/ψ control-region yields at 1 ab^{-1} .

The control region is essentially pure $J/\psi K^{*0}$ ($>99.8\%$) in both flavors, confirming it can be used, as in the published analyses, to calibrate reconstruction/PID efficiency and the $M_{bc}/\Delta E$ shape of the peaking background directly from data rather than from simulation.

6 Results

Multiplying the final cutflow efficiencies (Section 4) by the total expected rate at 1 ab^{-1} (Section 2) gives the expected yields:

flavor	S (signal)	B (residual $J/\psi K^{*0}$)	$S/\sqrt{S+B}$	rel. stat. unc. $\sqrt{S+B}/S$
e^+e^-	285.7	43.2	15.7	6.3%
$\mu^+\mu^-$	251.8	0.0 (0/2500 MC; 68% CL UL ~ 24.6)	15.9 (25.2 UL)	6.3% (6.6% UL)
combined	537.5	43.2	22.3	4.5%

Table 11: Expected yields and statistical precision at 1 ab^{-1} .

With zero background MC events surviving the veto in the muon channel, the point estimate $B_{\mu\mu} = 0$ is quoted together with a simple 68%-CL Poisson upper limit (1.14 raw events, scaled by the muon-channel weight) to avoid overstating precision from a single MC sample with finite statistics; a production analysis would use a much larger $J/\psi K^{*0}$ MC sample to pin this down (see Discussion).

At 1 ab^{-1} the study projects order-500 selected signal candidates split roughly evenly between the two lepton flavors, each with about 6% relative statistical precision on its own, and about 4.5% combined, neglecting the background not modeled here (continuum, generic $B\bar{B}$).

This is comparable in scale to the $\sim 40\text{--}80$ signal candidates reported by Belle II with 189 fb^{-1} (arXiv:2206.05946 [2]), scaled up by the larger 1 ab^{-1} sample size.

7 Discussion

Limitations, ordered by expected impact on a real measurement:

1. **Continuum ($e^+e^- \rightarrow q\bar{q}$) and generic $B\bar{B}$ combinatorial background are not modeled.** In every published $B \rightarrow K^*\ell\ell$ analysis these are the leading backgrounds, suppressed with continuum-suppression variables (Fox–Wolfram R_2 , event-shape/BDT discriminants) and fit with an ARGUS shape in M_{bc} plus a linear shape in ΔE (Wei et al.; Belle II arXiv:2206.05946 uses a 2D unbinned $M_{bc}\text{--}\Delta E$ fit rather than pure cut & count for this reason). A real analysis would add $q\bar{q}$ and generic $B\bar{B}$ MC samples, retrain the $K^*/M_{bc}/\Delta E$ cuts jointly with a continuum suppression cut, and extract the yield from a fit rather than a simple counting experiment. This is the single largest omission and likely dominates the true precision at 1 ab^{-1} .
2. **$\psi(2S)$ veto is not modeled** (no $B^0 \rightarrow \psi(2S)K^{*0}$ sample was generated), even though it is part of the standard published selection at a similar $m_{\ell\ell}$ offset from its resonance. Its expected rate is smaller than $J/\psi K^{*0}$ ($\mathcal{B}(\psi(2S) \rightarrow \ell\ell)$ is smaller and $\mathcal{B}(B \rightarrow \psi(2S)K^*)$ is also smaller) but should be added for completeness.
3. **Only one $J/\psi K^{*0}$ MC sample size (5000 events)** was available; the muon-channel veto efficiency is limited to “0/2500 survive”, i.e. a background estimate with essentially no MC statistics behind it. A production study needs a much larger charmonium-background MC sample (or the data control region itself, as in Section 5) to quote a reliable central value rather than an upper limit.
4. **No systematic uncertainties were evaluated** (tracking $\sim 0.3\%$ /track, lepton PID $\sim 1\text{--}2\%$ /lepton, K^*/kaon PID, $N_{B\bar{B}}/f_{00} \sim 1\text{--}2\%$, sub-decay BFs, all quoted at the level given by Wei et al. and the Belle II analyses above). At the $\sim 5\text{--}7\%$ statistical precision found here, PID and tracking systematics (few %) would already be a relevant, not negligible, contribution and should be added in quadrature.
5. **Electron bremsstrahlung is treated only implicitly** through the fast-simulation smearing and the wider ΔE /veto windows; no explicit photon-recovery (adding back a bremsstrahlung photon within a cone of the electron track) is modeled, unlike in the published analyses. This likely means the electron-channel efficiency and $m_{\ell\ell}$ resolution near the J/ψ edge are optimistic relative to real data.
6. **The K^* mass window was optimized on samples that both contain a genuine $K^*(892)$** , so it could not be tuned against combinatorial background as in a real analysis; its value ($0.896 \pm 0.075\text{ GeV}$) is a reasonable resolution-driven choice but should be re-optimized once combinatorial background is modeled.

8 Conclusion

Following the selection strategy of Wei et al. [Belle], PRL 103, 171801 (2009) [1] and the Belle II $B \rightarrow K^*(892)\ell\ell$ branching-fraction measurement (arXiv:2206.05946 [2]) — a K^* mass window,

M_{bc} and ΔE requirements, and asymmetric per-flavor J/ψ vetoes in $m_{\ell\ell}$ — this study projects, from existing signal and $J/\psi K^{*0}$ background ntuples reweighted to 1 ab^{-1} : about 286 e^+e^- and 252 $\mu^+\mu^-$ signal candidates, with a residual $J/\psi K^{*0}$ peaking background of about 43 events in the electron channel and consistent with zero (upper limit ~ 25 events) in the muon channel. The resulting statistical precision is about 6% per flavor and 4.5% combined, neglecting continuum and generic- $B\bar{B}$ combinatorial background, which are not modeled in this study and would need dedicated MC samples, a continuum-suppression cut, and a 2D M_{bc} - ΔE fit (as in the cited Belle and Belle II analyses) for a publication-grade result. The $J/\psi K^{*0}$ control region is confirmed to be $>99.8\%$ pure in both flavors, supporting its use, as in the published analyses, to calibrate efficiencies directly from data.

References

- [1] J.-T. Wei et al. (Belle Collaboration), Phys. Rev. Lett. **103**, 171801 (2009).
- [2] Belle II Collaboration, arXiv:2206.05946.

Figures

- Fig. 1 (01_mvisible_presel.png) — $K\pi$ mass, K^* window motivation.
- Fig. 2 (02_mbc_afterkst.png) — M_{bc} after the K^* window, $M_{bc} > 5.27\text{ GeV}$ cut.
- Figs. 3, 4 (03_deltae_ee.png, 03_deltae_mumu.png) — ΔE per flavor, asymmetric window motivation.
- Figs. 5, 6 (04_mll_ee_veto.png, 04_mll_mumu_veto.png) — dilepton mass, J/ψ veto per flavor.